Role of Incretin Hormones in Type 2 Diabetes Pathophysiology

Abstract

Incretin hormones, primarily glucagon-like peptide-1 (GLP-1) and glucose-dependent insulinotropic polypeptide (GIP), are gut peptides secreted in response to nutrient ingestion. They stimulate insulin secretion from pancreatic beta cells in a glucose-dependent manner. In patients with type 2 diabetes, the incretin effect is markedly blunted. Therapeutic strategies enhancing GLP-1 receptor activation or inhibiting dipeptidyl peptidase-4 (DPP-4) improve glycaemic control, promote weight loss, and confer cardioprotective benefits.

Introduction

Type 2 diabetes mellitus is a chronic metabolic disorder characterised by progressive beta-cell dysfunction and insulin resistance. Under physiological conditions, oral glucose administration elicits a significantly greater insulin response than intravenous glucose infusion. This phenomenon is termed the incretin effect.

Key Findings

Clinical trials have demonstrated that GLP-1 receptor agonists lower HbA1c levels by 1.0% to 1.8% and induce significant weight loss in overweight or obese patients. Furthermore, cardiovascular outcome trials confirmed reduced rates of major adverse cardiovascular events (MACE).

Limitations

Abstract-level data cannot substitute for comprehensive clinical trial reports. Potential gastrointestinal adverse events including nausea and vomiting were observed in approximately 15% to 30% of participants during initial titration.